

The stimulus-response chain

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

A stimulus-response chain describes how an organism reacts to a stimulus. At the beginning there is always a stimulus, that is, a change that can be detected by a sensory organ. An adequate stimulus is exactly the stimulus for which a sensory organ is especially specialised: the eye reacts mainly to light, the ear to sound and the nose to odours. The sensory organ detects the stimulus and converts it into electrical signals that can be transmitted in the nervous system. This information reaches the brain or central nervous system via afferent nerves. Afferent therefore means that information is conducted from a sensory organ toward the centre. In the brain, the incoming information is consciously processed, compared with existing knowledge and interpreted. Only there is it decided which reaction is useful. In this mode, the brain or CNS is therefore the place of conscious processing and control. The brain then sends a command via efferent nerves to an effector. Efferent means that the information travels away from the centre to an executing organ. An effector is the organ that actually carries out the reaction, for example a muscle. The observable reaction is finally the result of this control.

Task 1: Two examples of stimulus-response chains

Create two examples of a stimulus-response chain. Export both examples and insert them below.

Example 1 - insert here

Example 2 - insert here

Task 2: Description using technical terms

Describe the sequence of one of the two stimulus-response chains using technical terms.
